Skills Summary

Sequences, Series, and the Binomial Theorem

Use this reference while completing your worksheet or when studying.

Skill 1 — Evaluating a binomial coefficient

Definition: $\displaystyle\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$

It counts **unordered** selections of r things from n. Cancel the larger factorial in the denominator against the tail of $n!$ before multiplying, and remember the symmetry $\binom{n}{r} = \binom{n}{n-r}$.

Example: Evaluate $\displaystyle\binom{8}{3}$.

Step 1. Order does not matter in a binomial coefficient, so use the factorial definition and cancel before multiplying.

Step 2. $\displaystyle\frac{8!}{3!\,5!} = \frac{8\cdot 7\cdot 6}{3\cdot 2\cdot 1} = \frac{336}{6} = \mathbf{56}$.

Try it: Evaluate $\displaystyle\binom{10}{4}$.

check: **210**

Skill 2 — Expanding a binomial power in full

Binomial theorem:

$$(p+q)^n = \sum_{r=0}^{n} \binom{n}{r} p^{\,n-r} q^{\,r}$$

The two exponents always add to n. Whatever stands in the p or q slot goes there **whole** — a coefficient or a minus sign inside the bracket is raised to the power too.

Example: Expand $(x+2)^4$.

Step 1. A full expansion of a small power is fastest straight from the row of coefficients, taken with descending powers of x.

Step 2. Row 4 is $1, 4, 6, 4, 1$, and the powers of 2 climb as the powers of x fall: x^4, $4x^3(2)$, $6x^2(4)$, $4x(8)$, 16.

$\Rightarrow \quad \boldsymbol{x^4 + 8x^3 + 24x^2 + 32x + 16}$

Try it: Expand $(y-3)^3$.

check: $\boldsymbol{y^3 - 9y^2 + 27y - 27}$

Skill 3 — Extracting one specific term

General term: $T_{r+1} = \binom{n}{r} p^{n-r} q^{r}$

Method: match the exponent you were asked for to r (or to $n-r$), then evaluate *both* pieces — the binomial coefficient and the power of the constant — and multiply. Never expand the whole binomial for one term.

Example: Find the coefficient of x^3 in $(x+2)^5$.
Step 1. Only one term is wanted, so match the exponent to r instead of expanding.
Step 2. $x^{5-r} = x^3$ gives $r = 2$; then $\binom{5}{2} = 10$ and $2^2 = 4$, so the coefficient is $10 \cdot 4 = \mathbf{40}$.

Try it: Find the coefficient of x^2 in $(x+3)^4$.

check: **54**

Skill 4 — Reading the structure of a row

Two substitutions worth memorising:

- $p = q = 1$: $\sum_{r=0}^{n} \binom{n}{r} = 2^n$ — each row of Pascal's triangle sums to a power of two.
- $x = 1$: the sum of the coefficients of any polynomial is its value at $x = 1$, so $(1+cx)^n$ has coefficient sum $(1+c)^n$.

Example: Find the sum of the entries in row 5 of Pascal's triangle.
Step 1. A whole row summed is the $p = q = 1$ case of the binomial theorem, so no entry has to be listed.
Step 2. $\sum_{r=0}^{5} \binom{5}{r} = 2^5 = \mathbf{32}$; the entries $1+5+10+10+5+1$ confirm it.

Try it: Find the sum of the coefficients of $(1+3x)^4$.

check: **256**